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Review article

Applications of Palynology in Forensic Investigations: A Systematic Review

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Abstract

Forensic science, a multidisciplinary subject, encompasses the knowledge of physics, chemistry, geology, computer science, medical science, botany, zoology and molecular biology. Forensics is constantly renewed by adopting latest methods, procedures and techniques for solving legal issues. The study of pollen is one of the novel approaches of this aspect of science. This overview describes the significance of palynology, its applications, and different recent approaches and how it could be employed in solving criminal investigations.

Keywords

Botany, Molecular Biology, Forensics, Pollen, Palynology

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INTRODUCTION

Often, after a criminal incident, culprit escapes from the scene leaving little or no evidence for an investigation agent to correlate the incidence with available artifacts, drugs and food items recovered from the sight. In addition to this, culprit may deliberately generate some confusing stains, signs or artificial marks to make investigation far more difficult. The members of expert team which scrutinize objects (like - shoes, toys, papers, clothes), substances (including dirt particles, drug samples) chemicals (such as paints, explosives, toxins) tissue traces (viz. blood, hair, skin) and impressions (viz. fingerprints or footprints) left at the crime scene, are called forensic scientists. In this situation if a palynologist is present in the team of forensic scientists or investigators the court cases or legal issues can be easily solved (Singh, 2018).

Forensic palynology, using pollen science to solve crime dates back to 1950s Austrian murder mystery. The suspect had to confess his crime because suspect's boots contained 20- million-year-old fossilized pollen. The pollen grains recovered from suspects' footwear served as testimony to solve the issue. Even though forensic palynology has been successfully used as a crime fighting tool for over 50 years

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it is still not universally used in either civil or criminal cases except in a few countries (Mildenhall, 2008).

HIGHLY CONSERVED POLLEN FEATURES SERVE AS IMPORTANT TOOL IN FORENSICS

Pollen grains are very small structures formed inside male reproductive organ of a flower called anther. The pollen or pollen grain is a highly reduced male gametophyte, and is generally shed in desiccated condition. These tiny structures have only a very limited number of variable parameters (viz. the pollen wall with its structural and ornamental features) but show an astonishing diversity in few characters. A pollen grain represents a whole haploid generation in itself. Unlike the diploid plant, the haploid generation does not suffer from the selective pressure. Having no selective pressure pollen features remain preserved for millions of years. One of the conspicuous structural features of pollen is the surface ornamentation of exine, which is consistent for a given taxon (Shivanna and Johri, 1985). The peculiarity of pollen morphology and very small size of these structures make them a very useful tool in forensic investigations.

WHAT IS PALYNOLOGY?

Palynology is the science which deals with study of pollen and spores, in both living and fossil forms. Term palynology was first suggested by H. Hyde and P. Williams in 1944. However, an extensive study of pollen architecture was initially undertaken by Wodehouse and Erdtman. In India Dr. P.K.K. Nair is respectfully called as father of Indian Palynology owing to his major research contribution in palynology with special emphasis on the new principles, pollen morphology in relation to plant taxonomy and the triphyletic origin of angiosperms (Arti Garg, 2015). From decades, study of pollen grains and spores (either fresh or fossilized) has been the task of botanists or paleobotanists but now the study of these plant-propagules are used at least in few countries as an established tool for investigating criminal or civil court cases.

WHAT FORENSIC PALYNOLOGY IS?

Forensic palynology is the utilization of pollen and spores in solving legal issues, either civil or criminal (Bryant, 2013). In its broader application, the field of forensic palynology also includes legal information derived from the analysis of a broad range of organic-walled microscopic organisms-such as dinoflagellates, acritarchs, and ctinizoans-that can be found in both fresh and marine environments (Faegri et al, 1989). In much broader sense palynomorphs are defined as any organic-walled micro-fossil between 5 and 500 micrometers in size. Palynomorphs may be composed of organic materials such as chitin, pseudochitin, sporopollenin and dinosprin. Typically, palynomorphs are dinoflagellate cysts, acritarchs, spores, pollen, fungi, scolecodonts (scleroprotein teeth, jaws and associated features of plychaete, annelid worms), arthropod organs (such as insect mouth parts), chitinozoans and microforams. (Bakrac, 2015) These palynomorphs are helpful in pinpointing a smuggled item like antique pieces, drugs, smacks, packaged food etc. In addition to this, desired clues can be obtained from pollen grains inadvertently stuck on the substances or items like shoes, bags, clothes, electronics, weapons, blood, slime etc. recovered from the site where rape or assassination has occurred. In different genera of plants time of development and dispersal of pollen grains has been found different which is also helpful in correlating the actual time occurrence of any incidence. Certain plants are endemic or grow in a particular area which gives indication about the location at which an incident has occurred.

UNIQUE SPECALITIES OF POLLEN AND SPORES WHICH MAKE THEM USEFUL FORENSIC TOOLS

Diverse kind of plants grows in different parts of the globe. Due to unique macro- and micro-climatic condition specific kind of cryptogams and higher plants grows at any place whether it is land or water. As we know pollen is the male sex cell produced within anthers which in turn produce male gametes

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at the time of fertilization. Pollen grains are so small (ranging from 20 to 60 m in diameter) that they cannot be discerned by naked eye. The pollen grains or spores are, indeed, invisible observer for human being with an unaided eye because these structures can be discerned only with the aid of microscope. Apart from very small size pollen wall is variously sculptured in different species of plants. Not only wall-ornamentation but also the presence of a particular kind of aperture (germ pores) is the feature of a pollen grain. Also, pollen grains belong to which plant, growing in which geographical area is very important. Thus a pollen grain can be characterized by having a unique palynomorph just like a person has a unique fingerprint. A pollen fingerprint or a palynomorph is the number and type of pollen grains found in a geographical area at a particular month of a year (B.D. Mali, 2015). The using of pollen and spores in forensic sector lies in their combination of dispersal mechanisms, enormous production, microscopic size, species-specific wall ornamentation, highly recalcitrant nature to chemical or enzymatic action or mechanical abrasion or high temperature. The geological literature has documented that pollen can prevail for many thousands or millions of years, and it is generally found in relation to charcoal and other evidence of normal fires (Morales-Molino et al, 2012; Sniderman and Haberle, 2012). Actually the exine layer of pollen wall is composed of sporopollenin, an organic polymer of incredibly high stability. Sporopollenin is believed to be produced by oxidative polymerization of carotenoid pigments and carotenoid esters (Brooks and Shaw, 1971). Due to species-specific nature, parent plants can be identified collecting the tiny pollens and studying their details in the laboratory. They mostly fall within 2 km of the source and in many cases within 100 meters. Certain plants disperse their pollen merely within a few meters. On account of unique morphology, microscopic size, highly durable wall layers in drastic environments, property of sticking to any object due to variously sculptured pollen wall, pollen grains serve as excellent tools in forensic investigations.

On account of species-specific morphological characters, studying the pollen grains we can trace the parent plant from which pollen grains are shed. Most plants produce pollen grains in abundance which travel in the air not more than two kilometers of the source and in many cases within the range of 100 meters. Certain plants disperse their pollen merely within few meters.

CASES WHERE PALYNOLOGY HAS BEEN SUCCESSFULLY USED AS FORENSIC TOOL

There are few very interesting cases where science of palynology has been successfully used in solving the problem and catching the criminal. A few are cited as below.

Palynology was used as one of the major tools in war crime investigations in Bosnia. In July 1955, civilians were massacred and their bodies were buried in seven mass graves, following the fall of Srebrenica. Three months later the bodies were exhumed and transported to a number of new burial sites to hide the evidence of massacre and deflect the blame. However, palynological investigations clearly showed that the pollen and spores from the original mass graves had been transferred along with the bodies to the secondary burial site (Mali B.D., 2015).

During 1960s and 1970s, a noted Swiss criminalist, Max Frei, often used pollen as forensic tool to link suspect to events or to crime scenes (Palenik, 1982). In a well known case a suspect claimed that his pistol could not have been used to commit a recent murder because it had not been removed from its storage box in months. However, Dr. Frei probed the suspect was lying because grease on the pistol contained alder and birch pollen, both of which were pollinating when the murder occurred, not when the suspect claimed he had last cleaned the pistol and put it away.

In a rape case Horrocks and Walsh (1999) compared the pollen collected from the rape victim's clothing and the suspect's clothing with soil samples taken from alleged crime scene and the alibi scene. The two scenes were only seven meters apart but could be differentiated on the basis of pollen. The rape victim's clothing and the suspects clothing showed very close correlation with each other and the crime scene in the amounts of pollen types present. This supported strongly the presence of the suspect at the scene of crime.

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The above cases indicate the significance of palynology in forensics. When the visible clues are insignificant the tiny pollen which is too small to be seen by necked eye gives crystal clear indication in the direction of which investigation can be perused. Collection and analysis of palynomorphs is main skill of scientist. The collected material may be sediment, hair, cloths, illegal drugs or other artifacts. A forensic palynologist knows how to collect contamination-free samples and what precautions should be taken to ensure that the samples remain contamination-free throughout the storage, laboratory, and extraction phase and analysis process. Sufficient material also needs to be collected as in most cases the evidential material is very small in comparison to the comparator material.

After collection, the collected material is subjected to analysis. The collected material such as cloth-piece, soil, or mud can be washed with alcohol and pollen grains are sorted. Similarly hair of the suspect or victim can also be washed with detergent or distilled water and pollen grains are carefully separated. After repeated washing and acetolysis, the pollen are mounted on slide in glycerin. The comparator sample is extracted in the same manner. The comparator sample should match the evidential sample as closely as possible. For this both the slides are viewed through a high resolution microscope. The will then provide conclusive evidence of a connection between the suspect and the crime scene.

FUTURE PROSPECTS

In the present era of molecular biology, identification of pollen through DNA metabarcoding has a promising future. DNA metabarcoding could very likely increase its application to a broader range of situations (Bell et al 2016). Metagenomics is another modern technique which is used in medical and forensic investigations (Ghosh et al 2018). Forensic palynology along with novel molecular technique like DNA metabarcoding could be helpful in solving many legal issues but all that is required is the popularization this technique in the society and proper attention and interest of legal experts and scientists.

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